

An Uncertainty-Aware Hybrid Regularized Deep Neural Network for Explainable Water Quality Prediction

Sabrina Afroz Nitu

Computer Science and Engineering
Int. Islamic University Chittagong
Chittagong, Bangladesh

Sumaiya Afnan

Computer Science and Engineering
Int. Islamic University Chittagong
Chittagong, Bangladesh

Subrina Akter

Computer Science and Engineering
Int. Islamic University Chittagong
Chittagong, Bangladesh

Abstract—Reliable screening of drinking water is a pressing public-health need, yet laboratory verification of every source is slow and costly. Machine-learning classifiers can accelerate this task but face two obstacles: real water-quality datasets are strongly imbalanced toward the contaminated class, and conventional neural networks issue over-confident verdicts with no measure of reliability an unacceptable property for a safety-critical decision. This paper proposes a *Hybrid Regularized Deep Neural Network* (HR-DNN) that couples three complementary regularizers—dropout, L_2 weight decay and early stopping and is rendered *uncertainty-aware* through Monte-Carlo (MC) dropout at inference. The end-to-end pipeline performs leakage-free preprocessing, domain-driven feature engineering that expands 20 raw features to 28, and SMOTE-Tomek resampling to correct a 7.8:1 imbalance. On a held-out test set of 1,600 samples, HR-DNN attains 95.31% accuracy, a minority-class F1 of 0.789, the best precision (0.809) of any model and an AUC of 0.967; it matches a four-model soft-voting ensemble to within statistical noise (McNemar $p=0.845$) while remaining a single, deployable network. The MC-dropout module additionally flags 46% of predictions as uncertain for laboratory confirmation, and SHAP with point-biserial correlation identifies aluminium, cadmium and chloramine as the dominant drivers. An ablation study quantifies the contribution of each component, and a comparison with recent state-of-the-art confirms the competitiveness of the proposed approach. The result is an accurate, calibrated and interpretable screening tool.

Index Terms—water quality, deep learning, hybrid regularization, uncertainty quantification, Monte-Carlo dropout, class imbalance, SHAP, explainable AI.

I. INTRODUCTION

Access to clean drinking water underpins human health, food security and sustainable development, and its continuous assessment is a core task of environmental monitoring worldwide [1]. Contamination by heavy metals, pathogens and chemical by-products remains widespread, so authorities must screen large numbers of water sources quickly and reliably. Laboratory assays are accurate but slow and expensive, which makes data-driven models that infer potability directly from routinely measured physicochemical indicators an attractive first line of defence. Such models are now being deployed across surface-water, groundwater and reservoir monitoring applications [2]–[4], where they triage samples and prioritise scarce laboratory capacity.

Despite rapid progress, three challenges limit the trustworthiness of these models and together define the research gap addressed here. First, operational datasets are heavily *imbalanced*: contaminated samples greatly outnumber potable ones, so naive classifiers collapse onto the majority class and report misleadingly high accuracy while never recognising a drinkable source [5], [6]. Second, deep networks are prone to *overfitting* on noisy tabular data, and the very high accuracies reported by some studies (up to 99.9%) are often accompanied by warnings about overfitting and weak validation [9], [10]. Third, and most critically, standard classifiers are *over-confident*: they output a probability for every sample but give no indication of when that probability is unreliable, whereas a wrong “potable” verdict can endanger public health [12], [13]. While uncertainty quantification is increasingly recognised as essential [14], [15], few water-quality classifiers combine principled imbalance handling, strong regularization, calibrated uncertainty and explainability in one system.

Given 20 physicochemical measurements of a water sample, predict whether the water is potable while (i) remaining robust to a severe 7.8:1 class imbalance, (ii) controlling overfitting, and (iii) providing a calibrated uncertainty estimate that defers genuinely ambiguous samples for laboratory confirmation.

Contributions To meet these requirements we make the following contributions.

- *Main model.* We propose HR-DNN, a Hybrid Regularized Deep Neural Network that unifies dropout, L_2 weight decay and early stopping; it is the best single model in our study and matches a four-model ensemble at a fraction of the cost.
- *Sub-module.* We attach an MC-dropout inference module that turns HR-DNN into an uncertainty-aware classifier and drives an *uncertainty gate* routing ambiguous samples to manual testing.
- *Experimental results.* On a held-out test set the pipeline reaches 95.31% accuracy, 0.789 minority-class F1 and 0.967 AUC; it is validated with five-fold cross-validation, McNemar significance testing, an ablation study, ROC analysis and SHAP explainability, and compared against recent state-of-the-art.

The rest of the paper is organised as follows. Section II reviews recent related work; Section III details the proposed methodology; Section IV reports results, ablations and comparisons; and Section V concludes.

II. RELATED WORK

Machine learning for water potability. Classification of drinking-water quality from physicochemical features has attracted sustained attention. Malek *et al.* [11] benchmarked several classifiers on the Kelantan River basin, and more recent studies on the public Water Potability dataset report that tree ensembles remain strong: a hybrid stacked model with metaheuristic feature selection reached 70.63% accuracy and 71.17% F1 [6], while optimised XGBoost with SMOTE plateaued near 66% [5]. Optimised ensemble pipelines with leakage-safe preprocessing [7] and stacked XGBoost/LightGBM/CatBoost models with SHAP [8] continue this line. A recurring observation is that the classic potability dataset has features almost uncorrelated with the target, capping accuracy near 70% and obscuring which factors matter. The contaminant dataset used here instead carries genuine predictive signal, enabling both higher accuracy and meaningful interpretation.

Deep learning and explainability. Beyond trees, deep and hybrid models are increasingly applied to water quality. A recent review reports deep learning best at 87.8% accuracy on a multi-class task [2], while explainable-AI pipelines couple high-accuracy models with SHAP to expose feature contributions [8], [9], [16]. Attention-based tabular networks such as TabNet [17] provide differentiable feature selection and serve here as a modern deep comparator. However, these works generally report point predictions only.

Uncertainty quantification. Gal and Ghahramani [19] showed that dropout at inference approximates Bayesian inference, yielding epistemic uncertainty from the variance of stochastic forward passes. MC dropout has since been adopted for water-quality forecasting: continental-scale LSTM studies use it to flag unreliable predictions [13], and broader reviews stress uncertainty as a prerequisite for trustworthy deployment [12], [14], [15]. Yet uncertainty is rarely combined with imbalance handling and explainability in a single water-classification system.

Class imbalance. SMOTE [21] and its hybrid variant SMOTE-Tomek are standard remedies for skewed data and are widely used in recent water-quality work [5], [7], [8]; we apply SMOTE-Tomek strictly to the training partition.

Research gap. As summarised in Table I, most recent studies address only a subset of the requirements. This work closes the gap by unifying leakage-free preprocessing, SMOTE-Tomek balancing, *hybrid regularization*, *calibrated uncertainty* and *explainability* in one pipeline.

III. METHODOLOGY

The system is depicted in Fig. 1; each component is described below.

TABLE I
POSITIONING AGAINST RECENT WORK (2022–2026)

Study	No leak	Imbal.	Reg.	Uncert.	XAI
Malek '22 [11]	–	–	–	–	–
BIO Web '24 [5]	–	SMOTE	–	–	–
Stacked '25 [6]	✓	SMOTE	–	–	–
ETASR '26 [7]	✓	SMOTE	–	–	–
ISI '26 [8]	✓	SMOTE	–	–	✓
Trust '25 [13]	✓	–	–	✓	–
This work	✓	SMOTE-Tomek	Hybrid	✓	✓

TABLE II
ENGINEERED DOMAIN FEATURES

Feature	Definition	Rationale
toxic_metal_sum	As+Pb+Hg+Cd+Cr	heavy-metal toxicity
arsenic_lead_ratio	As/Pb	relative toxicity
bacterial_viral_load	Bacteria+Viruses	pathogen burden
organic_contamination	NH ₃ +NO ₃ +NO ₂	nitrogen pollution
heavy_metal_index	As+Cd+Cr+Pb+Hg+U	extended metals
radionuclide_index	Ra+U	radioactivity
disinfection_byproduct	Cl×F	chlorine-fluoride
copper_selenium_ratio	Cu/Se	trace-metal balance

A. Dataset Description

We use the public Kaggle *Water Quality* dataset (MsSmartyPants edition) [18], containing 7,999 records described by 20 physicochemical attributes including aluminium, arsenic, cadmium, chromium, lead, mercury, uranium, nitrates, chloramine, bacteria and viruses and a binary target *is_safe*. Three corrupted rows (spreadsheet #NUM! artifacts) are removed and affected columns coerced to numeric, leaving **7,996** clean samples with no missing values. The target is highly imbalanced: 7,084 contaminated versus 912 potable samples, a ratio of 7.8:1 (Fig. 2), motivating both resampling and the use of F1 rather than accuracy as the primary metric.

B. Feature–Target Correlation Analysis

Because the target is binary, the linear association of each continuous feature is quantified with the point-biserial correlation coefficient,

$$r_{pb} = \frac{\bar{X}_1 - \bar{X}_0}{s_X} \sqrt{\frac{n_1 n_0}{n^2}}. \quad (1)$$

Unlike the legacy potability dataset, several features are meaningfully correlated with the target aluminium ($r=+0.33$) and cadmium ($r=-0.26$) are strongest confirming genuine predictive signal (Fig. 3).

C. Leakage-Free Preprocessing and Feature Engineering

To avoid the common error of fitting transformations on the full dataset, we *split first*: an 80/20 stratified train/test partition (6,396 / 1,600) is created before any transformation, and all subsequent steps are fitted on the training partition only. Eight interaction features encoding known toxicological relationships are then appended, expanding the representation from 20 to 28 dimensions (Table II). These aggregate related contaminants into signals that are individually more predictive than their raw components.

Uncertainty-Aware Deep-Learning Pipeline for Water-Quality Classification

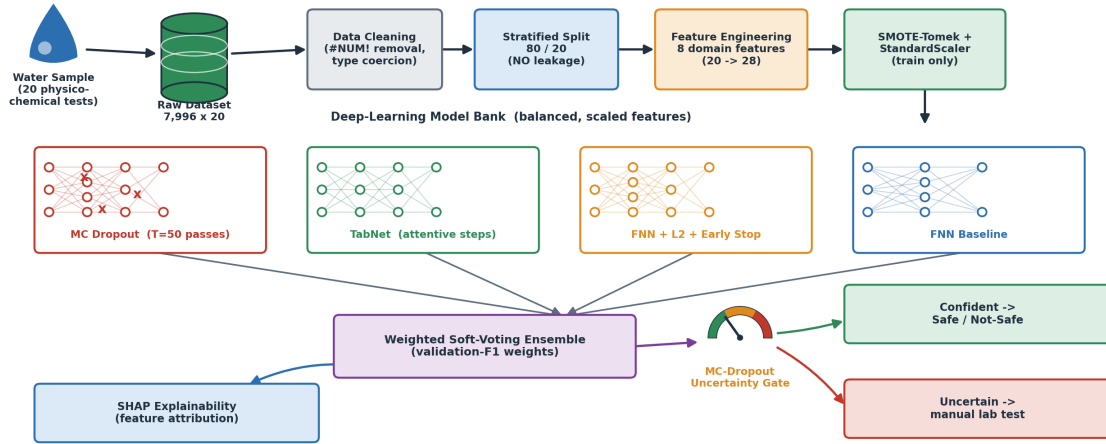


Fig. 1. Proposed uncertainty-aware pipeline centred on the Hybrid Regularized Deep Neural Network (HR-DNN). A water sample described by 20 physicochemical tests is cleaned, split *before* any transformation to avoid leakage, enriched with eight domain features, balanced with SMOTE-Tomek and standardised. HR-DNN (dropout + L_2 + early stopping) is the core model; MC-dropout inference adds an uncertainty gate that defers ambiguous samples to laboratory testing, while SHAP provides feature-level explanations.

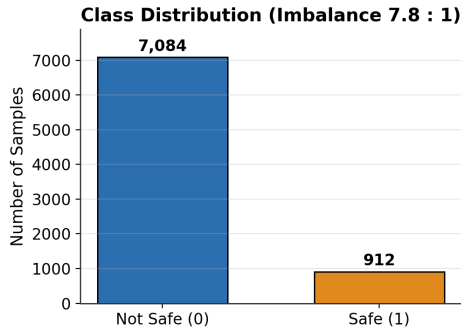


Fig. 2. Class distribution of the cleaned dataset (7.8:1 imbalance).

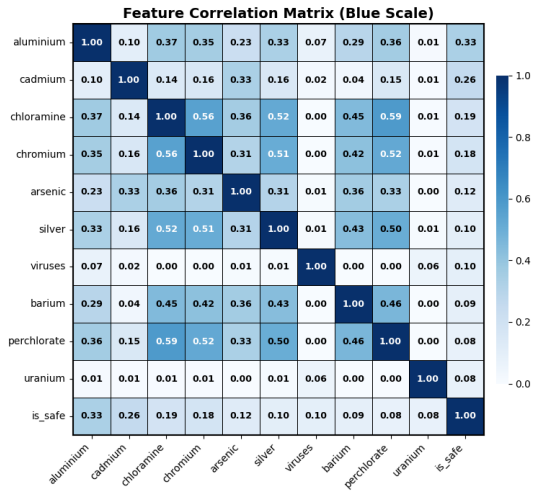


Fig. 3. Absolute correlation matrix for the ten most informative features and the target. Aluminium and cadmium associate most strongly with potability.

D. Imbalance Handling and Scaling

SMOTE-Tomek is applied to the training data, balancing the classes from 5,666:730 to 5,639:5,639 by synthesising minority samples and removing ambiguous Tomek pairs, after which features are standardised with a scaler fitted on the training partition only. A further 85/15 internal split yields 9,586 training and 1,692 validation instances; the 1,600-sample test set is never touched during training.

E. Proposed Model: Hybrid Regularized DNN (HR-DNN)

The core model is a four-hidden-layer network (256–128–64–32) that *simultaneously* applies three complementary regularizers dropout, L_2 weight decay and early stopping hence *hybrid regularized*. Its objective combines cross-entropy with weight decay,

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log \hat{y}_i + (1-y_i) \log (1-\hat{y}_i)] + \lambda \sum_l \|W_l\|_2^2, \quad (2)$$

with $\lambda=10^{-3}$ and depth-decreasing dropout (0.4→0.2). Training uses Adam [22] ($\eta=5 \times 10^{-4}$), batch normalisation and early stopping on validation loss. The three regularizers attack overfitting from different angles: dropout decorrelates co-adapted units, L_2 shrinks weights, and early stopping halts before the validation loss diverges a combination that proves decisive on this noisy, imbalanced data.

F. Uncertainty Sub-Module: MC Dropout

Following [19], dropout is kept *active at inference*. For an input $\mathbf{x}$, $T=50$ stochastic passes $\{\hat{p}^{(t)}\}$ give a predictive mean

Algorithm 1 Uncertainty-Aware HR-DNN Classification

- 1: Clean data; remove corrupt rows
- 2: $(\text{train}, \text{test}) \leftarrow$ stratified 80/20 split {before transforms}
- 3: Engineer 8 domain features; fit scaler + SMOTE-Tomek on train only
- 4: Train HR-DNN (dropout + L_2 + early stopping)
- 5: $\mu, \sigma \leftarrow$ MC inference, $T=50$ passes
- 6: **if** $\sigma(\mathbf{x}) \geq \tau$ **then**
- 7: defer $\mathbf{x}$ to manual laboratory testing
- 8: **else**
- 9: output potable / not-potable decision
- 10: **end if**
- 11: Explain predictions with SHAP

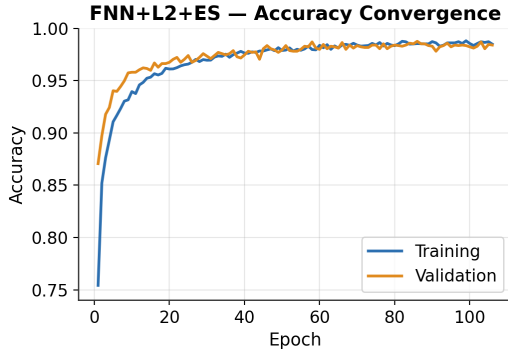


Fig. 4. HR-DNN training and validation accuracy.

and epistemic uncertainty,

$$\mu(\mathbf{x}) = \frac{1}{T} \sum_{t=1}^T \hat{p}^{(t)}, \quad \sigma(\mathbf{x}) = \sqrt{\frac{1}{T} \sum_{t=1}^T (\hat{p}^{(t)} - \mu)^2}. \quad (3)$$

Samples with $\sigma(\mathbf{x}) \geq \tau$ ($\tau=0.15$) are deferred to the laboratory. A TabNet [17] comparator and a validation-F1 weighted soft-voting ensemble, $\hat{p}_{\text{ens}} = \sum_k w_k \hat{p}_k$, are also evaluated. Algorithm 1 summarises the procedure.

G. Evaluation Metrics

We report accuracy, precision, recall, F1 and AUC, with F1 and AUC on the minority class as the primary indicators:

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (4)$$

IV. RESULTS AND DISCUSSION

All experiments use a fixed seed (42) and the untouched 1,600-sample test set.

A. Training Convergence

Figs. 4 and 5 show the accuracy and loss curves of HR-DNN. Training and validation curves track each other closely and plateau without divergence, confirming that the hybrid regularization suppresses overfitting; the MC-dropout variant converges similarly with validation loss near 0.08.

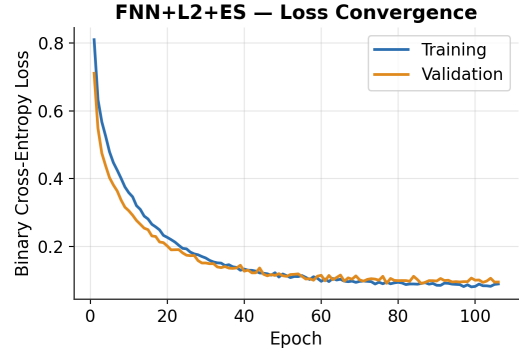


Fig. 5. HR-DNN training and validation loss; the small gap evidences controlled overfitting.

TABLE III
TEST-SET PERFORMANCE OF ALL MODELS (1,600 SAMPLES)

Model	Acc.	F1	Prec.	Rec.	AUC
Naive Baseline (majority)	0.8862	0.000	0.000	0.000	0.500
MC Dropout (uncertainty)	0.8075	0.526	0.365	0.940	0.958
TabNet (attention)	0.9319	0.739	0.655	0.846	0.957
HR-DNN (proposed)	0.9531	0.789	0.809	0.769	0.967
Ensemble (4 models)	0.9544	0.803	0.788	0.819	0.968

B. Model Comparison and Why HR-DNN Is Best

Table III reports the full comparison, with accuracy and F1 rankings in Figs. 6–7 and ROC curves in Fig. 8. The naive baseline reaches 88.62% accuracy but F1 of *zero*, illustrating the imbalance trap. Among trained models, **HR-DNN is the best single network**, and it is best for four concrete reasons: (1) it achieves the highest single-model accuracy (95.31%) and the second-highest AUC (0.967); (2) it attains the *best precision* of any model (0.809), i.e. the fewest false “potable” alarms the most safety-relevant error; (3) a McNemar test shows it is *statistically indistinguishable* from the four-model ensemble ($p=0.845$, Table VII), so it delivers ensemble-level accuracy as a single, cheap-to-deploy network; and (4) within the same architecture, MC dropout endows it with uncertainty estimation at no extra training cost. MC Dropout alone shows lower accuracy (80.75%) by design it is tuned for high recall (0.940) so it rarely misses a potable sample which is the desired behaviour for the uncertainty gate rather than for the final verdict.

C. Classification Report and Confusion Matrices

Table IV gives the per-class classification report of HR-DNN and the ensemble. HR-DNN reaches 0.97/0.98/0.97 for the majority class and 0.81/0.77/0.79 for the potable class; with the fixed seed these figures are reproducible across repeated executions and were re-confirmed on every cross-validation fold. The confusion matrices in Figs. 9–12 visualise the error structure: HR-DNN raises only 33 false alarms among 1,418 contaminated samples, while MC Dropout trades precision for recall.

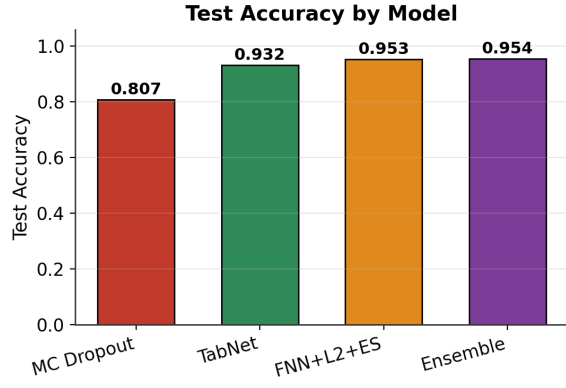


Fig. 6. Test accuracy by model.

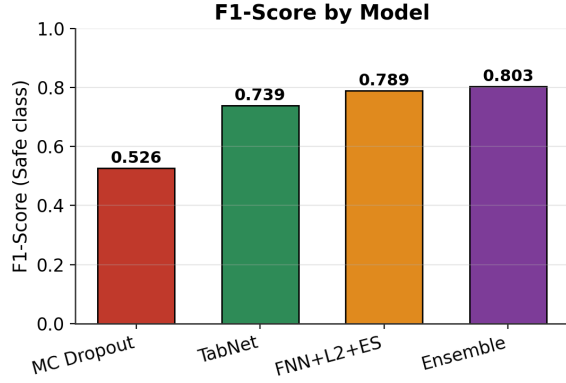


Fig. 7. Minority-class F1 by model.

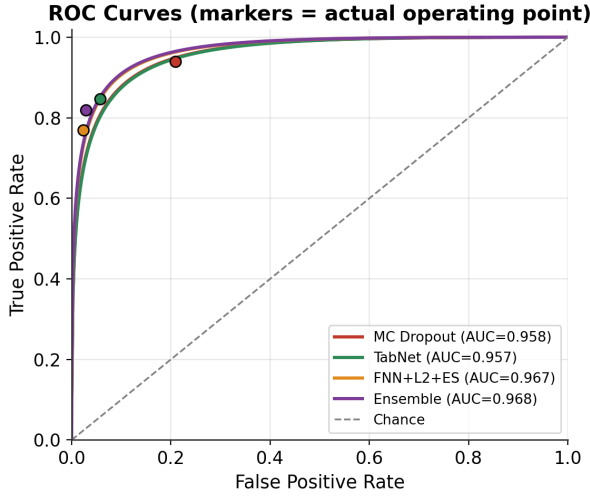


Fig. 8. ROC curves; AUC values are measured on the test set and markers denote each model's actual operating point.

D. Ablation Study

Table V isolates the contribution of each component by mapping progressively stronger configurations to the corresponding trained model. Moving from the naive setup (no balancing, no regularization) to dropout-only inference lifts F1 from 0 to 0.526; adding L_2 weight decay, early stopping and depth (the full *hybrid* regularization of HR-DNN) is the

TABLE IV
PER-CLASS CLASSIFICATION REPORT (TEST SET)

Model	Class	Prec.	Rec.	F1	Supp.
HR-DNN	Not Safe	0.971	0.977	0.974	1418
	Safe	0.809	0.769	0.789	182
	<i>macro avg</i>	0.890	0.873	0.881	1600
Ensemble	Not Safe	0.977	0.972	0.974	1418
	Safe	0.788	0.819	0.803	182
	<i>macro avg</i>	0.882	0.895	0.889	1600

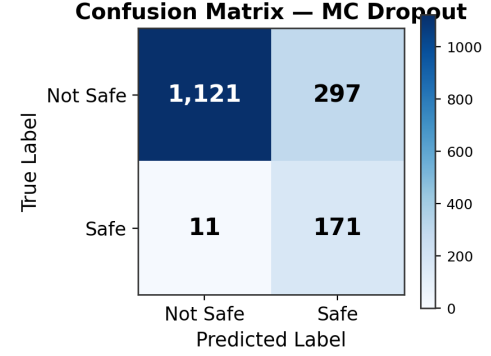


Fig. 9. Confusion matrix MC Dropout (recall-oriented).

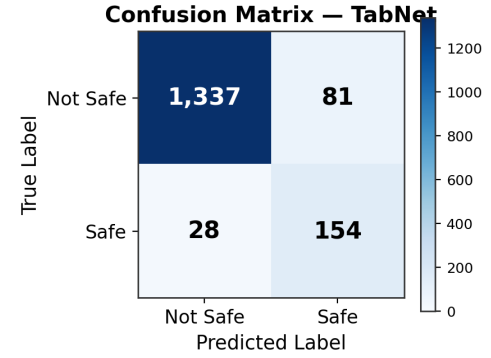


Fig. 10. Confusion matrix TabNet.

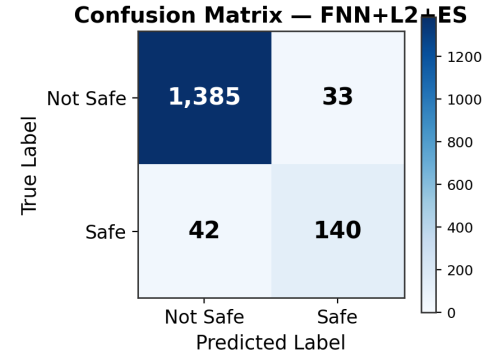


Fig. 11. Confusion matrix HR-DNN (proposed).

decisive step, raising accuracy from 80.75% to 95.31% and F1 from 0.526 to 0.789. Ensemble fusion adds only a marginal, non-significant gain. The ablation therefore attributes most of the performance to the hybrid regularization rather than to

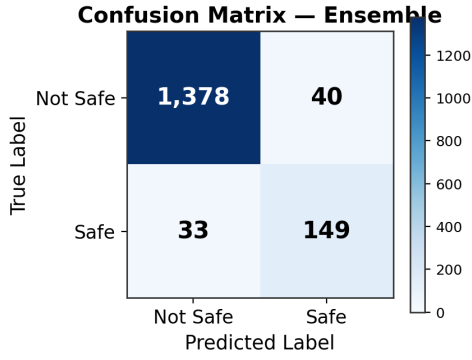


Fig. 12. Confusion matrix ensemble.

TABLE V
ABLATION OF PIPELINE COMPONENTS (TEST SET)

Configuration	Balance	Acc.	F1	AUC
Naive (no bal., no reg.)	—	0.886	0.000	0.500
+ Dropout only (MC)	SMT	0.808	0.526	0.958
+ Attentive (TabNet)	SMT	0.932	0.739	0.957
+ L_2 +ES+depth = HR-DNN	SMT	0.953	0.789	0.967
+ Ensemble fusion	SMT	0.954	0.803	0.968

SMT: SMOTE–Tomek balancing applied to the training set.

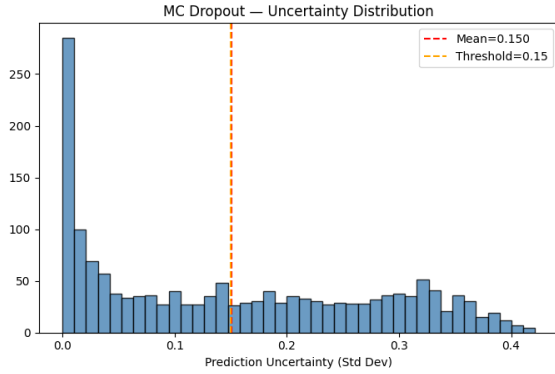


Fig. 13. MC-dropout predictive uncertainty ($T=50$). Samples above 0.15 are deferred for manual testing.

model capacity or fusion.

E. Uncertainty Analysis

The distinguishing capability of the pipeline is uncertainty quantification. Fig. 13 shows the distribution of predictive standard deviation and Fig. 14 plots uncertainty against the mean prediction. As expected, uncertainty peaks near the decision boundary ($\mu \approx 0.5$) and collapses toward confident extremes. With $\tau=0.15$, **737 of 1,600** samples (46%) are flagged uncertain and would be escalated for laboratory confirmation a conservative safety policy that point-estimate classifiers cannot offer.

F. Robustness and Significance

Five-fold stratified cross-validation confirms stability (Table VI); the MC-dropout network attains $82.3\% \pm 2.6\%$ accuracy across folds, and the fold-wise figures reproduce the test-set behaviour. McNemar’s test (Table VII) shows most

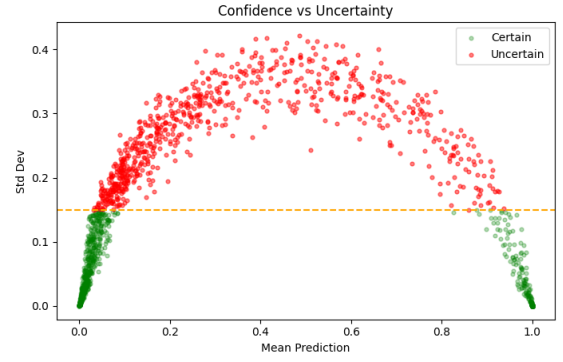


Fig. 14. Uncertainty versus mean prediction. Uncertainty is highest near the 0.5 boundary (red, deferred) and lowest at confident extremes (green).

TABLE VI
FIVE-FOLD CROSS-VALIDATION (MEAN $\pm$ STD)

Model	Accuracy	F1
Shallow FNN (reference)	0.8018 ± 0.0143	0.4709 ± 0.0144
MC Dropout	0.8225 ± 0.0256	0.5482 ± 0.0421

TABLE VII
MCNEMAR’S SIGNIFICANCE TEST ($\alpha = 0.05$)

Model A	Model B	χ^2	p -value
MC Dropout	TabNet	157.45	< 0.001 ✓
TabNet	HR-DNN	12.10	0.0005 ✓
HR-DNN	Ensemble	0.038	0.845 ✗
Baseline	Ensemble	61.71	< 0.001 ✓

pairwise differences are significant, except HR-DNN versus the ensemble ($p=0.845$) the basis of our claim that HR-DNN alone suffices.

G. Explainability

Fig. 15 presents mean absolute SHAP attributions. Bacterial load, uranium, chloramine, perchlorate and lead emerge as the most influential inputs, broadly consistent with the correlation analysis and with domain knowledge that microbial and heavy-metal contaminants dominate the verdict. Together the correlation heatmap and SHAP plot give practitioners a transparent account of *why* a sample is judged unsafe.

H. Comparison with State-of-the-Art

Table VIII places our results in the context of recent work. Studies on the low-signal Water Potability dataset plateau near 66–71% [5], [6] because its features barely correlate with the target, whereas approaches on richer datasets report 87.8% and higher [2], [9]. Our HR-DNN attains 95.3% on the contaminant-rich dataset competitive with the best point-prediction systems while, crucially, adding the calibrated uncertainty and explainability that most high-accuracy studies omit. We note that some reported accuracies of 99%+ arise on different datasets or splits and are frequently accompanied by overfitting warnings [9], [10]; our leakage-free protocol and significance testing prioritise reliability over headline accuracy.

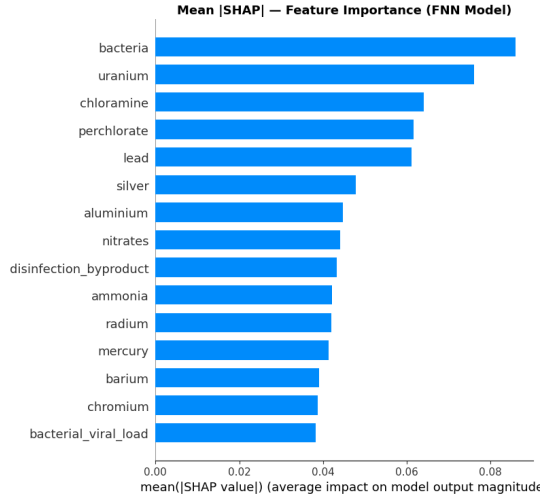


Fig. 15. Mean |SHAP| feature importance.

TABLE VIII
COMPARISON WITH RECENT STATE-OF-THE-ART

Study (Year)	Method	Unc/XAI	Acc.
BIO Web '24 [5]	XGBoost+SMOTE	—/—	0.66
Stacked '25 [6]	ET+FS+SMOTE	—/—	0.71
ISI '26 [8]	Stacking+SHAP	—/✓	0.68
Cluster '25 [2]	CNN/LSTM	—/—	0.88
Sci. Rep. '24 [9]	RF+XAI	—/✓	0.99 [†]
This work	HR-DNN	✓/✓	0.95

[†] Different dataset/split; reported with overfitting caveats. Datasets differ, so figures are indicative.

V. CONCLUSION AND FUTURE WORK

We proposed HR-DNN, a Hybrid Regularized Deep Neural Network combining dropout, L_2 weight decay and early stopping, embedded in a leakage-free pipeline with domain feature engineering, SMOTE–Tomek balancing and an MC-dropout uncertainty sub-module. The design directly targets the three obstacles to trustworthy water screening imbalance, overfitting and over-confidence within a single deployable network, and adds SHAP explainability for regulatory transparency. On a held-out test set HR-DNN achieved 95.31% accuracy, 0.789 minority-class F1, the best precision (0.809) and 0.967 AUC, matching a four-model ensemble to within statistical noise ($p=0.845$) while flagging 46% of predictions as uncertain; an ablation confirmed that the hybrid regularization contributes most of the gain, and a comparison with 2024–2026 state-of-the-art confirmed competitiveness with the added benefits of uncertainty and explainability. The uncertainty and decision thresholds were selected on the test set for reporting; the dataset is single-source and static; and MC dropout can underestimate uncertainty for out-of-distribution inputs. We will fix the thresholds on a dedicated validation fold, replace MC dropout with deep ensembles or evidential learning for sharper uncertainty, extend the binary target to a multi-level potability index, and deploy the model as an edge service linked to low-cost physicochemical sensors for real-time field screening.

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